

Technical Design Document (TDD): Cold Chain Control Tower

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<https://github.com/UjjwalK08/supply-chain-logisitics-FDE-project>

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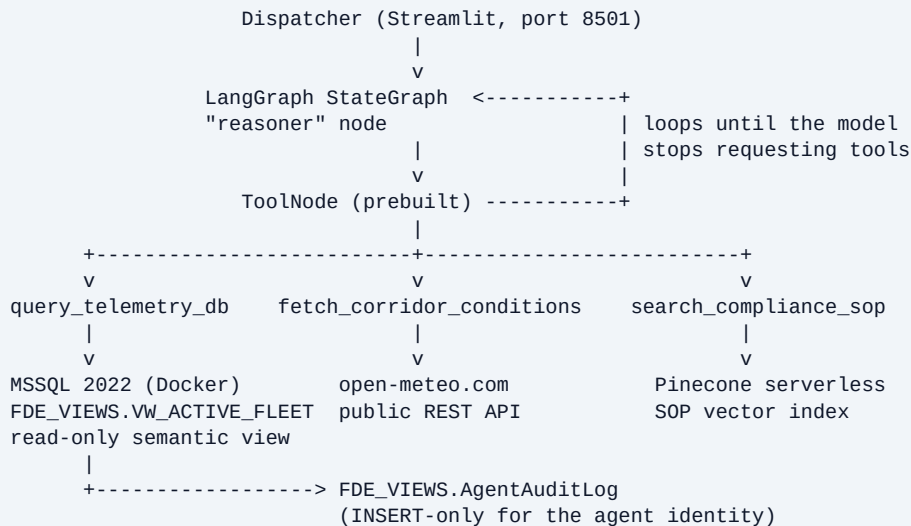
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1. System Architecture (HLD)

The Cold Chain Control Tower is a decoupled, three-tier platform that lets control-tower operators query fleet telemetry, environmental conditions, and compliance SOPs through a natural-language interface. It performs **Text-to-SQL** over a governed view, calls external APIs, and retrieves governing policy clauses, then synthesises the three into a cited action plan.

1.1 Architectural Overview

A stateless presentation layer, a state-machine orchestration engine, and a hardened data access layer.



1.2 Component Responsibilities

- **Presentation Tier** — Streamlit application (`src/ui.py`). Holds UI session state only: a per-browser thread identifier, rendered message history, and cached dashboard queries. All reasoning is delegated to the orchestration tier.
- **Orchestration Engine** — A LangGraph StateGraph (`src/orchestrator.py`). Manages the execution lifecycle: intent deconstruction, tool selection, iteration, and synthesis of tool output into a structured report. This is a hand-assembled graph, not `create_react_agent`; the two-node loop is explicit and inspectable.
- **Data & Integration Tier** — Decoupled backends. The agent is isolated from raw data by a database view and a least-privilege role, so tier boundaries are enforced by the database engine rather than by application code.

2. Agent Orchestration (LLD)

The orchestration tier constrains probabilistic behaviour with a deterministic state machine.

2.1 State Management & Execution Loop

State is a single messages list under LangGraph's `add_messages` reducer. The graph has two nodes and one conditional edge:

Element	Implementation
reasoner	Invokes the tool-bound LLM with the system prompt prepended at call time
tools	LangGraph's prebuilt <code>ToolNode</code> over the three tool functions
Routing	<code>tools_condition</code> — routes to <code>tools</code> if the reply contains tool calls, else to <code>END</code>
Persistence	MemorySaver checkpoint, keyed by <code>thread_id</code> (one per browser session)

Execution sequence:

- 1 **Ingestion** — The dispatcher's message is appended to graph state.

- 2 **Reasoning** — The reasoner sees the system prompt plus full conversation history and either emits tool calls or produces a final answer. Tool choice is the model's decision; nothing in the graph forces a particular tool or order.
- 3 **Tool execution** — `ToolNode` runs every requested call and appends one `ToolMessage` per `tool_call_id`.
- 4 **Iteration** — Control returns to the reasoner, which may call further tools using the results just returned. This is what allows multi-hop questions: a location query feeds coordinates into a weather lookup, whose output is then checked against the SOP.
- 5 **Synthesis** — When no further tools are needed, the reasoner emits the final report.

The system prompt is prepended inside the reasoner node rather than seeded into stored state. This guarantees every entry point (the Streamlit console and the CLI harness) is governed by the same contract, and keeps the prompt out of the persisted history.

2.2 Model Routing & Provider Independence

Both the reasoner and the embedding model are selected at runtime by environment variable, so the stack runs fully hosted or entirely local without code changes.

Variable	Values	Effect
<code>Agent_llm</code>	OPENAI / DEEPSEEK / anything else	GPT-4o, DeepSeek, or a local Ollama fallback
<code>Embeddings_model</code>	OPENAI / LOCAL	1536-dim hosted, or 1024-dim local BGE-M3

2.3 Resiliency & Fallbacks

- **Graceful degradation** — Each tool catches its own exceptions and returns a descriptive string rather than raising. A failed weather call or database error becomes context the model can reason about and report, instead of an unhandled crash.
- **Dashboard isolation** — The fleet snapshot powering the KPI tiles and map fails independently of the chat. If the database is unreachable the dashboard hides itself and surfaces the underlying error; the conversational path stays available.
- **Thread recovery** — If a run is interrupted between the reasoner and the tool node, the checkpoint retains an assistant message with tool calls but no matching tool results. Replaying that history is rejected by any OpenAI-compatible API, permanently poisoning the thread. The presentation tier detects this specific rejection, rotates to a fresh thread, and retries once. The chat input is additionally disabled while a run is in flight, which prevents the condition arising in the first place.

3. Data & Security Model (LLD)

Security is enforced at the database layer, so a prompt-injected or hallucinated destructive statement is rejected by the server rather than being filtered in Python.

3.1 Telemetry Security (VW_ACTIVE_FLEET)

The agent interacts solely with the read-only view `FDE_VIEWS.VW_ACTIVE_FLEET`. The `query_telemetry_db` tool generates T-SQL directly from the dispatcher's question; the view is what makes that Text-to-SQL step tractable, because it renames a deliberately legacy schema into readable English so the model can reason without a schema dictionary. Generation accuracy and blast radius are therefore controlled by the

same artefact.

View column	Underlying legacy column	Type
Timestamp	TS.UTC	datetime
Latitude	V_LAT	float
Longitude	V_LON	float
Current_Temperature_C	IOT_TEMP_VAL_C	float (explicitly cast)
Cargo_Condition_Code	CGO_COND_CD	float, 0–1 continuous
Risk_Classification	RISK_CLS_TXT	text
Delay_Probability	DELAY_PROB_DEC	float, 0–1
Port_Congestion_Level	PRT_CNG_LVL	float, 0–10
Route_Risk_Index	RT_RSK_IDX	float

Security principle: exposing only a flat, view-based projection means the agent has no name by which to reference the base table. Combined with the grants in 3.2, DROP, UPDATE and DELETE are rejected by the engine.

A secondary check in `query_telemetry_db` rejects any statement not beginning with SELECT. This is defence in depth for clearer error messages — it is explicitly **not** the security boundary, which lives in the grants below.

3.2 Role-Based Access Control (RBAC)

Connections use the `USR_FDE_R0` service account:

```
GRANT SELECT ON FDE_VIEWS.VW_ACTIVE_FLEET TO USR_FDE_R0;
DENY SELECT ON dbo.TBL_SC_FLEET_HIST_RAW TO USR_FDE_R0;
DENY INSERT, UPDATE, DELETE, ALTER ON SCHEMA::dbo TO USR_FDE_R0;
GRANT INSERT ON FDE_VIEWS.AgentAuditLog TO USR_FDE_R0;
```

The account cannot read the raw table, cannot modify schema, and cannot write anywhere except the audit table. The DENY on `SCHEMA::dbo` also covers tables added to that schema in future, so the boundary does not erode as the database grows.

Administrative credentials (`SQL_ADMIN_USERNAME`) are separate, used only by the ingestion script and the audit viewer, and are never given to the agent.

3.3 Auditing

Every decision cycle is written to `FDE_VIEWS.AgentAuditLog`:

Column	Type	Contents
LogID	INT IDENTITY	Primary key
Timestamp	DATETIME	Defaults to GETDATE()
SessionID	VARCHAR(50)	Browser thread identifier
NodeExecuted	VARCHAR(50)	reasoner, tools, or reasoner_final

Column	Type	Contents
ToolName	VARCHAR(100)	Tool invoked, or LLM Text Synthesis
Content	NVARCHAR(MAX)	JSON arguments, raw tool output, or final answer

One row is written per tool call, per tool result, and per final synthesis, giving a complete replay of how any answer was produced.

Scope of the guarantee: the log is append-only **with respect to the agent identity** — `USR_FDE_RO` holds `INSERT` and nothing else, so the agent cannot alter or erase its own trail. It is not cryptographically immutable: an administrator retains full rights over the table. Achieving true immutability would require a ledger table, append-only storage, or off-box log shipping. This is a deliberate scope decision, not an oversight.

4. Retrieval Layer (LLD)

4.1 Document Ingestion Pipeline

`scripts/ingest_sop_pinecone.py` maintains the SOP index incrementally:

- **Change detection** — Each source file is MD5-hashed against a cache. Unchanged files are skipped entirely.
- **Deterministic chunk identity** — Chunks are keyed `{filename}-chunk-{n}`, so a modified document is deleted and re-upserted cleanly rather than accumulating orphans.
- **Deletion propagation** — Files removed from `data/policy/` are purged from the index.
- **Format coverage** — Markdown (header-aware splitting), plain text, PDF, CSV and XLSX.
- **Dimension self-healing** — An index found at the wrong dimensionality is dropped and recreated before ingestion proceeds.

4.2 Index Isolation

Hosted and local embeddings write to separate indexes (`fde-sop-index-openai` at 1536 dimensions, `fde-sop-index-local` at 1024). Because the vector spaces are incompatible, isolating them means switching providers can never silently corrupt an existing index.

5. Presentation Tier

5.1 Workspaces

Workspace	Purpose
Dispatch console	Conversational interface, onboarding, live reasoning trace
Fleet overview	KPI tiles, geospatial scatter, underlying records
Security & audit	Admin-gated reader over the audit log

Reasoning is rendered transparently: each tool call's generated arguments and raw output are shown in expandable panels as they occur, and are retained under a collapsed panel beneath the answer in replayed history. Each answer carries badges for elapsed time, tool call count, and which tools fired.

5.2 Output Contract

The system prompt (src/prompts/system_prompt.txt) constrains every final answer to a fixed structure:

- 1 **Executive Summary** — the anomaly and the immediate operational risk
- 2 **Telemetry & Environment Analysis** — a table of readings, plus a causal reading of whether the environment or the equipment is at fault
- 3 **Required Action Plan** — numbered steps derived from the SOP, with an explicit citation of the document and rule driving them

This is what converts raw tool output into something a dispatcher can act on without interpreting JSON.

6. Deployment Strategy

6.1 Infrastructure Topology

Node	Contents
App node (EC2, Ubuntu)	Streamlit under a systemd unit with Restart=always
Database node (EC2, Ubuntu)	MSSQL 2022 in Docker, volume-backed at /var/opt/mssql
Managed services	Pinecone serverless (AWS us-east-1); the configured LLM provider

The ODBC Driver 18 for SQL Server and unixodbc-dev are prerequisites on the app node.

6.2 Network Perimeter Security

- 1 **Ingress (public)** — Only port 8501 on the app node is publicly reachable.
- 2 **Internal** — Port 1433 is restricted by security group rule to the private IP of the app node. The database is not addressable from the public internet.

7. Known Limitations

Stated explicitly so the design is not read as claiming more than it delivers.

Area	Current state
Self-correction	Not implemented. A failed query returns an error string; the model may retry of its own accord, but no retry loop is coded.
Clarification	No dedicated clarification node. An ambiguous request is answered on the model's own assumptions.
Human-in-the-loop	No approval or interrupt step. The agent is read-only and advisory, so it recommends escalations rather than performing them — but a reviewer should not infer a gated workflow exists.
Audit immutability	Append-only for the agent identity, not tamper-proof against an administrator. See 3.3.
Dataset realism	The dataset is historical (2021-01-01 to 2024-08-29), not a live feed, and its GPS coordinates are synthetic — generated uniformly within a lat 30–50 / lon -120 to -70 box and clipped to its edges. Roughly one point in eight falls over open water. Distances and routes are therefore not physically meaningful.
Semantic naming	Cargo_Condition_Code is a continuous 0–1 score despite the Code suffix, which may lead the model to treat it as categorical.

Area	Current state
Scale	Dashboard queries are capped at a 500-row snapshot with a 5-minute cache. No pagination or incremental refresh.